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BSIT – 4C

System Administration and Maintenance

BIOS vs UEFI Comparison

Feature	BIOS	UEFI
Definition	Basic Input/Output System; traditional firmware used to initialize hardware and start the operating system.	Unified Extensible Firmware Interface; modern firmware interface that replaces BIOS and provides advanced boot and system management features.
Boot Process	Uses a 16-bit boot process and loads the bootloader from the Master Boot Record (MBR).	Uses a 32-bit or 64-bit environment and loads boot files from the EFI System Partition (ESP).
Maximum Disk Support	Up to 2 TB.	More than 2 TB (supports very large drives).
Partition Style	MBR (Master Boot Record)	GPT (GUID Partition Table)
Security Features	Limited security; no Secure Boot support.	Supports Secure Boot, firmware validation, and better protection against boot-time malware.
Speed	Generally slower hardware initialization and boot process.	Faster startup and boot times due to optimized initialization and parallel device loading.
Advantages	Simple, widely supported on older hardware, and compatible with legacy operating systems.	Supports large disks, Secure Boot, graphical setup utilities, faster booting, and modern hardware features.
Disadvantages	Limited disk size support, fewer security features, slower boot process, and outdated architecture.	Slightly more complex configuration and possible compatibility issues with very old operating systems or hardware.
Modern Usage	Mostly found on older or legacy systems.	Standard firmware used in modern desktops, laptops, servers, and virtual machines.

Why UEFI Has Largely Replaced BIOS in Modern Computers

The traditional BIOS served as the standard firmware for personal computers for many years. Its main responsibility was to initialize hardware components such as the processor, memory, storage devices, and keyboard before transferring control to the operating system. Although BIOS was reliable for older computers, its design was based on 16-bit architecture and became increasingly limited as computer hardware evolved.

One major limitation of BIOS is its dependence on the Master Boot Record (MBR) partitioning scheme. MBR can only support disks up to 2 terabytes and allows a maximum of four primary partitions. As modern computers commonly use much larger solid-state drives and hard disks, these restrictions became impractical for enterprise and consumer systems.

UEFI (Unified Extensible Firmware Interface) was developed to overcome these limitations. Unlike BIOS, UEFI supports the GUID Partition Table (GPT), which allows very large storage devices and a much larger number of partitions. This makes UEFI more suitable for modern operating systems such as Ubuntu, Windows 11, and other current Linux distributions.

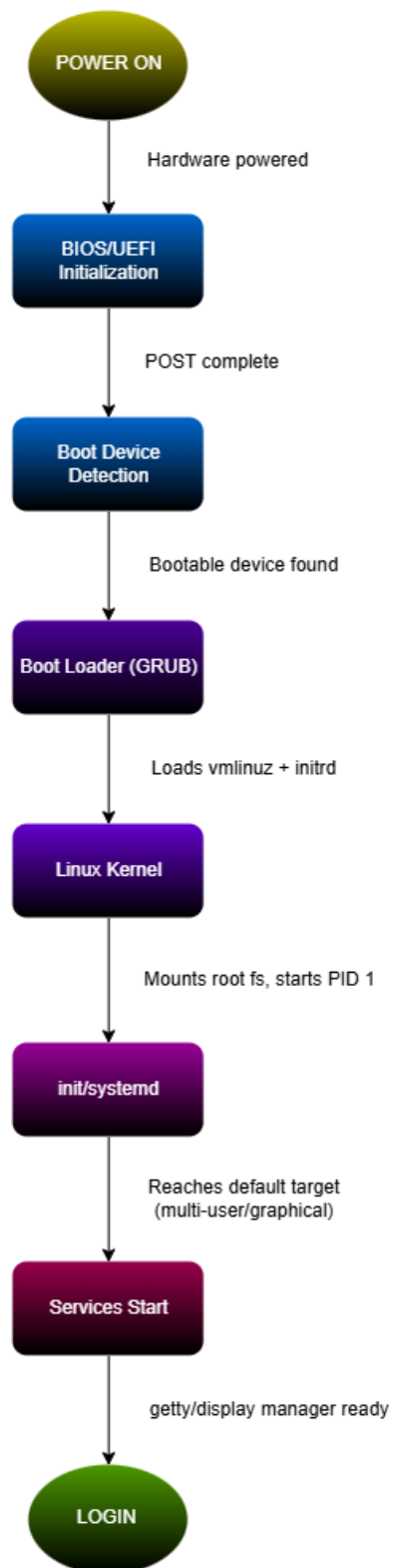
Another important reason UEFI has replaced BIOS is security. UEFI includes Secure Boot, which verifies the digital signature of bootloaders and operating system components before they are executed. This helps prevent rootkits and other malware from loading during the startup process, significantly improving system security.

UEFI also provides faster boot performance. It initializes hardware more efficiently and can load multiple components in parallel, reducing the time required for a computer to start. In addition, UEFI offers a graphical configuration interface, mouse support, network-based booting, and extensible firmware features that are not available in traditional BIOS.

In virtualized and enterprise environments, UEFI has become the preferred standard because it provides better compatibility with modern hardware, supports advanced storage technologies, improves security, and offers more flexible system management capabilities. For these reasons, most modern computers and virtual machines now use UEFI instead of BIOS, making it the dominant firmware standard for current computing systems.

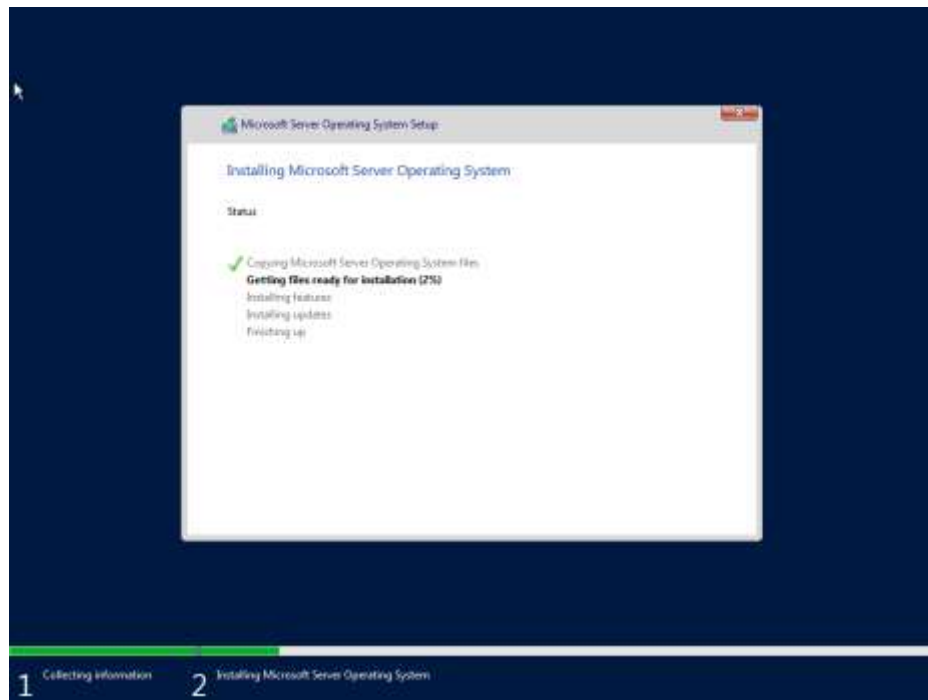
Boot Process Flowchart

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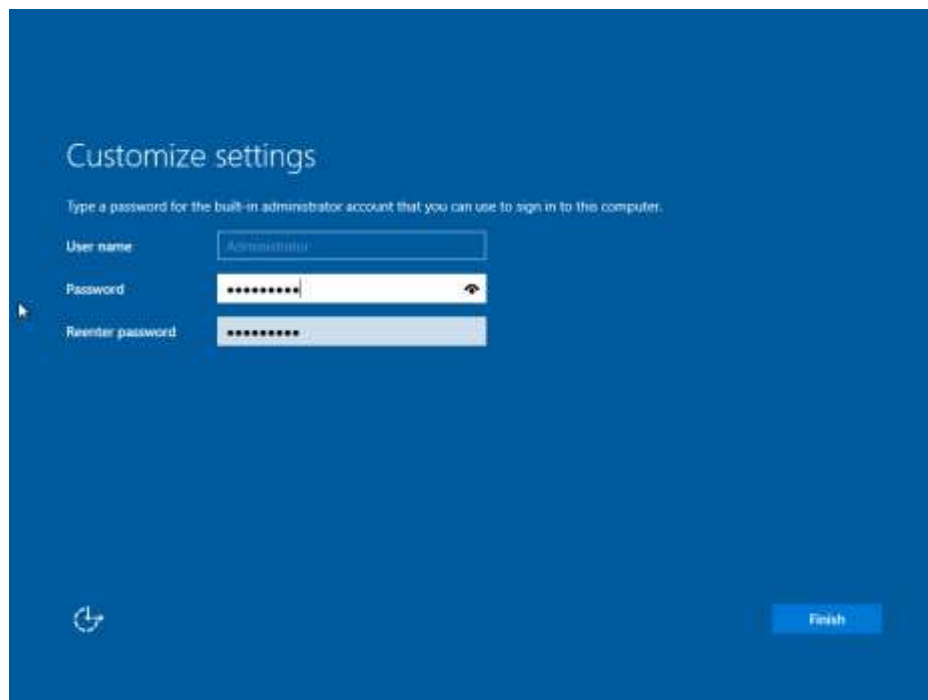


Install Windows Server Evaluation

Installation



Admin Pass



Successful Login



Comparison Report

Windows Server vs Ubuntu Server vs Rocky Linux

Feature	Windows Server	Ubuntu Server	Rocky Linux
Licensing	Commercial / Proprietary	Free and Open Source	Free and Open Source
User Interface	Graphical (GUI) and Server Core	Primarily Command Line (optional GUI)	Primarily Command Line (optional GUI)
Package Management	Windows Update, MSI, PowerShell, Winget	apt / dpkg	dnf / rpm
Security	Active Directory, Group Policy, Windows Defender, BitLocker	AppArmor, UFW, strong Linux permissions	SELinux, firewalld, enterprise-grade security

Performance	Higher resource usage due to GUI and Windows services	Lightweight and efficient	Very stable and optimized for server workloads
Typical Enterprise Use Cases	Active Directory, file servers, domain controllers, Microsoft application hosting	Web servers, cloud infrastructure, containers, DevOps platforms	Enterprise Linux servers, databases, virtualization, hosting environments
Advantages	Easy administration, excellent Microsoft ecosystem integration, strong GUI tools	Free, large community support, fast updates, excellent cloud support	Long-term stability, RHEL compatibility, predictable enterprise behavior
Disadvantages	Requires licensing, higher hardware requirements, more resource intensive	Steeper learning curve for GUI-focused users	Smaller community than Ubuntu, fewer beginner-oriented resources

Discussion:

Windows Server, Ubuntu Server, and Rocky Linux are widely used server operating systems, but they are designed for different administrative environments and enterprise requirements.

Windows Server is best suited for organizations that rely heavily on Microsoft technologies such as Active Directory, Group Policy, Microsoft SQL Server, Exchange Server, and .NET applications. Its graphical administration tools make it easier for beginners and administrators who prefer a GUI-based environment. However, Windows Server requires commercial licensing and generally consumes more system resources than Linux-based alternatives.

Ubuntu Server is one of the most popular Linux server distributions, especially in cloud computing and DevOps environments. It uses the APT package management system, provides extensive community documentation, and supports modern technologies such as Docker, Kubernetes, and OpenStack. Ubuntu Server is lightweight, free to use, and receives frequent

updates, making it an excellent choice for web servers, application hosting, and cloud infrastructure.

Rocky Linux is an enterprise-focused Linux distribution that is fully compatible with Red Hat Enterprise Linux (RHEL). It emphasizes long-term stability, predictable updates, and strong security through SELinux and firewalld. Rocky Linux is commonly used for enterprise databases, virtualization hosts, and production servers where reliability is more important than having the latest software versions.

In terms of security, Linux distributions generally benefit from strong permission models and open-source transparency, while Windows Server provides centralized security management through Active Directory and Group Policy. Regarding performance, Ubuntu Server and Rocky Linux typically require fewer resources than Windows Server, allowing them to run efficiently on lower-end hardware or in virtualized environments.

Overall, Windows Server is ideal for Microsoft-centered enterprise networks, Ubuntu Server is ideal for cloud-native and web-based deployments, and Rocky Linux is ideal for stable enterprise Linux infrastructures requiring RHEL compatibility. Understanding the strengths and limitations of each operating system helps system administrators choose the most appropriate platform for specific organizational and technical requirements.